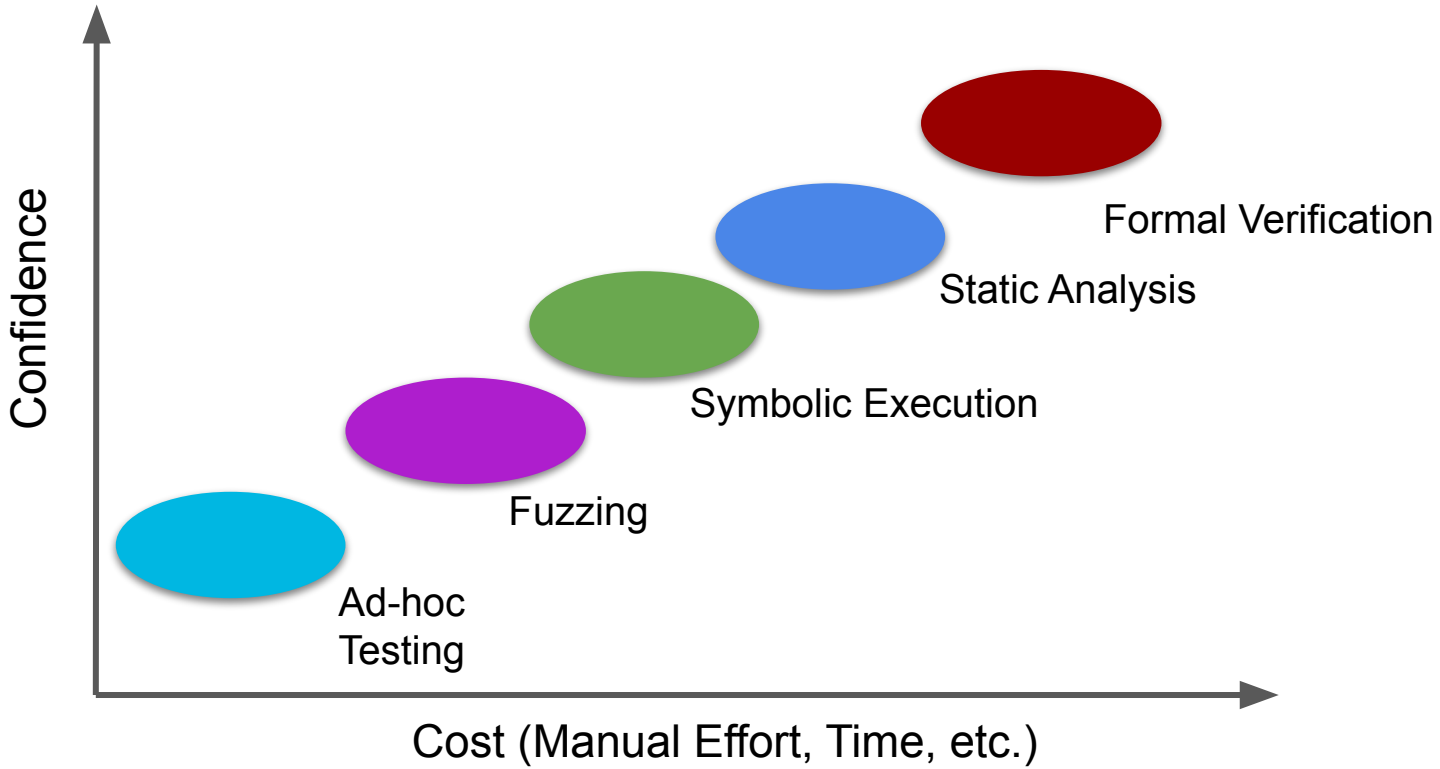
Symbolic Execution

Announcements

- No class 4/21 and 4/30
- Lab8 - due last night
- HW3 - due tonight
- Project part 1 - Resubmission deadline 4/13 at midnight
  - Part 2 deadline extended to 4/21

Landscape of Program Analysis Techniques



Motivation

How would the existing techniques we've seen attempt to find this bug?

1. Randoop
2. EvoSuite
3. AFL
4. Formal Verification - SAT solving
5. Formal Verification - deductive reasoning

```
public static boolean foo(String[] args) {
    int a = Integer.parseInt(args[0]);
    int b = Integer.parseInt(args[1]);

    boolean flag;

    if (a > 10 && b < 100) {
        int x = 2 * a + 3 * b;
        int y = 5 * a - b;

        if (x == 245 && y == 111) {
            System.out.println("You triggered the secret logic!");
            flag = true; //bug
        } else {
            System.out.println("Not quite the magic numbers.");
        }

    } else {
        System.out.println("Input out of range.");
    }

  return flag;
}
```

Symbolic Execution

- A middle ground between random testing and formal verification
- Treats the program as *symbols* rather than concrete values
- Instead of running the program with specific inputs like `x = 5`, use a placeholder value `x = α` where alpha is a placeholder representing any value
- As the program “executes”, keep track of the symbolic expression for each value

Concrete Execution

```
assume (N >= 0);
X := 0;
Y := 1; //P1
while X < N do {
   X := X + 1;
   Y := Y * X;
}
assert (Y = N!);
```

Inputs are concrete values

- A concrete state: maps from variables to concrete values

  when N=3, and after P1, we have the state

  {X=0, Y=1, N=3}

- Execution of a program statement
  - Go from an input concrete state to an output concrete state
  - "`X=X+1`" goes from state {X=0, Y=1, N=3} to {X=1,Y=1, N=3}

Concrete Execution

JVM holds LVT values for

`a, b, flag, x, y, args`

During execution of each program statement, the LVTs are updated.

Requires concrete inputs for args

```
public static boolean foo(String[] args) {
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  return flag;
}
```

Symbolic Execution

Inputs are represented symbolically

$\alpha 1, \alpha 2, \alpha 3$, …

- Variables get symbolic values
- A symbolic value is either a:

  constant (e.g., an integer constant),

  symbol (αi)

  expression formed from αi and constants ($\alpha 1 + \alpha 2, 3\alpha 3$)

Symbolic Execution

args = [α1, α2]

```
public static boolean foo(String[] args) {
    int a = Integer.parseInt(args[0]);
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    boolean flag;

    if (a > 10 && b < 100) {
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}
```

Symbolic States

A symbolic state consists of:

- A variable state:
  - Map from variable to symbolic values
  - {X: $2\alpha1 + 3\alpha2$, Y: $5\alpha1 - \alpha2$ }

- A path condition (PC):
  - A boolean condition that must hold when the program reaches this point
  - $(2\alpha1 + 3\alpha2 = 245) \wedge (5\alpha1 - \alpha2 = 111)$

Symbolic Execution Rules

For each program statement, we need to define a “rule” for how the values get updated

Similar to deductive reasoning rules

Before and after each rule statement we have a **variable state (VS)** and a **path condition (PC)**

Notation

VS_e = variable state at the entry of statement S

VS_x = variable state at the exit of statement S

PC_e = path condition at the entry of statement S

PC_x = path condition at the exit of statement S

Init: every input variable is assigned a symbol and PC = True

Assignment (X := E)

$VS_x = VS_e [X \rightarrow VS_e(E)]$

$PC_x = PC_e$

The new value of X is the symbolic value of E

Path condition is unchanged

A Simple Example

// input variables: A,B,X,Y,Z

{A:α1, B:α2, X:α3, Y:α4, Z:α5}, True

X := A + B;

{A:α1, B:α2, X:α1+α2, Y:α4, Z:α5} , True

Y := A - B;

{A:α1, B:α2, X:α1+α2, Y:α1-α2, Z:α5} , True

Z := X + Y

{A:α1, B:α2, X:α1+α2, Y:α1-α2, Z:(α1+α2)+(α1-α2)} , True

{A:α1, B:α2, X:α1+α2, Y:α1-α2, Z: 2α1} , True

Assume B

- Variable state unchanged

$$VS_x = VS_e$$

- Path condition adds the assumption

$$PC_x = PC_e \wedge VS_e(B)$$

Assert B

- If PC_e implies $VS_e(B)$

  $VS_x = VS_e$

  $PC_x = PC_e$

- If PCe does not imply VSe(B)

  print “assertion failed“

  terminate the evaluation

Example

//Inputs A, B, X, Y, and Z

{A:α1, B:α2, X:α3, Y:α4, Z:α5}, True

assume (A>B);

{A:α1, B:α2, X:α3, Y:α4, Z:α5}, $\alpha\alpha$1>$\alpha\alpha$2

X := A + B;

{A:α1, B:α2, X:α1+α2, Y:α4, Z:α5} , α1>α2

Y := A - B;

{A:α1, B:α2, X:α1+α2, Y:α1-α2, Z:α5} , α1>α2

Z := X + Y

{A:α1, B:α2, X:α1+α2, Y:α1-α2, Z:(α1+α2)+(α1-α2)} , α1>α2

assert (X=A+B $\wedge$ Y=A-B $\wedge$ Z=2*A $\wedge$ Y>0);

α1>α2 $\rightarrow$ (α1+α2 = α1+α2 $\wedge$ α1-α2 = α1-α2 $\wedge$ α1+α2 +α1-α2 = 2α1 $\wedge$ α1-α2>0) ???

Exercise

//inputs A, B, X

assume (A=2*B)

X := A + B;

X := X - B;

X := X - 2*B;

assert (X=0)

Conditionals

If B then S1 else S2

Three cases:

1. PC_e -> VS_e (B)
2. PC_e -> ! VS_e (B)
3. Path condition does not imply either the if or else branch

Conditionals

If B then S1 else S2

Case one: Current path condition implies the if condition

1. $PC_e \rightarrow VS_e(B)$

Add the if-condition to our path condition

$PC_x = PC_e \wedge VS_e(B)$

Variable state remains unchanged

$VS_x = VS_e$

Conditionals

If B then S1 else S2

Case two: Current path condition implies the else condition

1. PC_e -> $!VS_e(B)$

Add the **negation** if-condition to our path condition

$PC_x = PC_e \wedge VS_e(B)$

Variable state remains unchanged

$VS_x = VS_e$

Loops

Example with a ton of path conditions

Show how much things grow

Number of paths = 2^n

Suddenly we have a ton of calls to the SAT solver....

Path explosion problem!

What might be difficult to model symbolically?

```
BufferedReader reader = new BufferedReader(new FileReader("input.txt"));

String line = reader.readLine();
```

Parameterized Unit Tests (PUTs)

Unit tests where the inputs are left as symbols and

Example here

Lab Today: JavaPathFinder

Lab Today

Running a symbolic executor on a benchmark and observing the outputs

Summary

- Symbolic execution
  - Ahdsjfsaldjfa

- Next class we discuss ways to tackle the path explosion problem